# Sec 8.2 (Series)

What is a series? A series is a sum of the terms of a sequence.

Ex. Consider the sequence $a_n = \frac{1}{2^n}$ The series for this sequence would be defined as

$$\sum_{n=0}^{N} a_n = \sum_{n=0}^{N} \frac{1}{2^n} = \frac{1}{2^0} + \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^4} + \cdots + \frac{1}{2^N}$$

We can also have infinite series where we add infinitely many terms:

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} \frac{1}{2^n} = \frac{1}{2^0} + \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^4} + \cdots$$

Ex Consider sequence $a_n = n^2 + 1$ Write the sum of the first 10 terms in series notation. What is the sum of the first 10 terms?